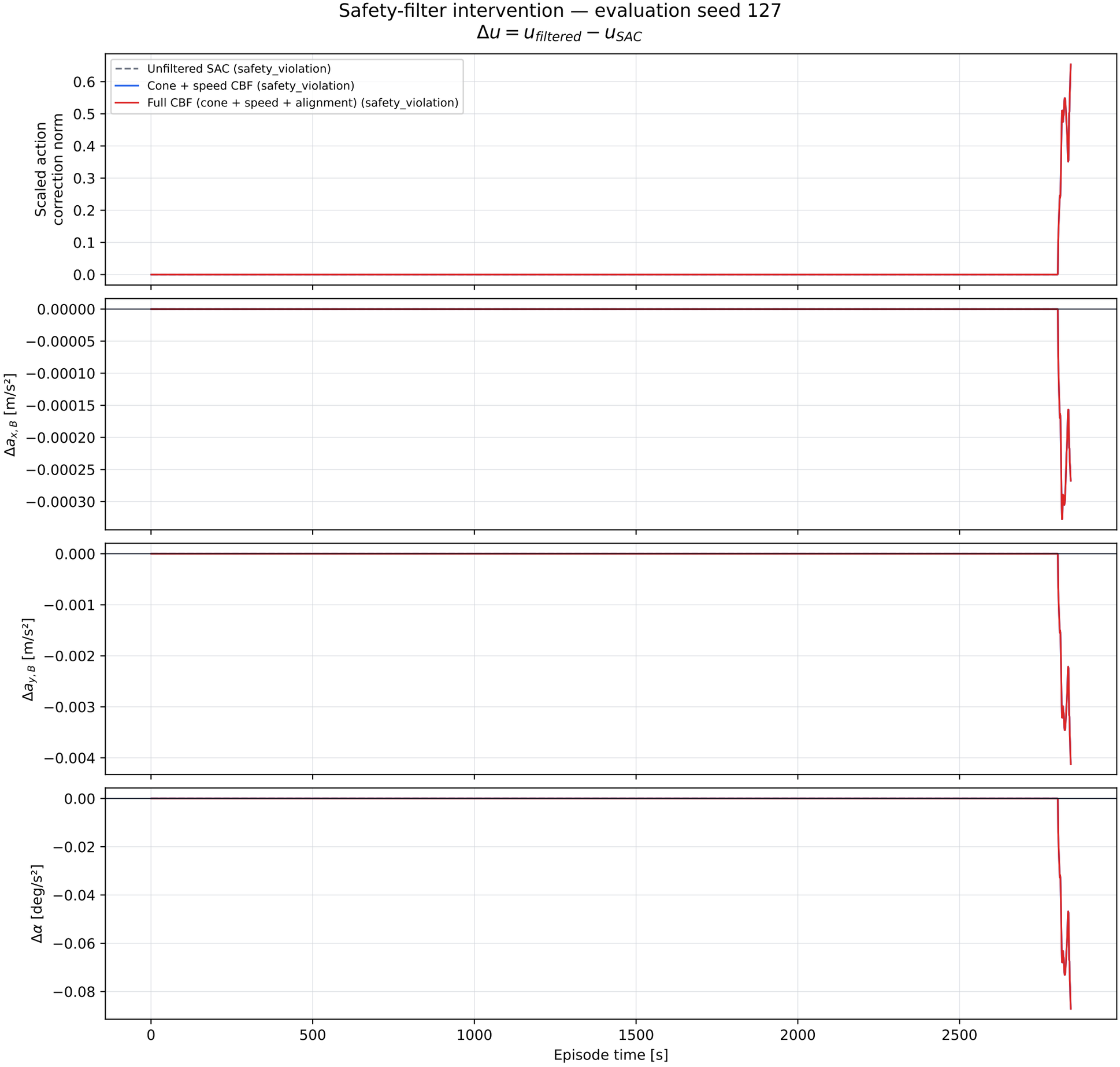
Safety-filter intervention — evaluation seed 127
$\Delta u = u_{filtered} - u_{SAC}$
Unfiltered SAC (safety_violation)
Cone + speed CBF (safety_violation)
Full CBF (cone + speed + alignment) (safety_violation)
Scaled action correction norm
0.0
0.1
0.2
0.3
0.4
0.5
0.6
$\Delta a_{x,B}$ [m/s²]
0.00000
−0.00005
−0.00010
−0.00015
−0.00020
−0.00025
−0.00030
$\Delta a_{y,B}$ [m/s²]
0.000
−0.001
−0.002
−0.003
−0.004
$\Delta\alpha$ [deg/s²]
0.00
−0.02
−0.04
−0.06
−0.08
0
500
1000
1500
2000
2500
Episode time [s]